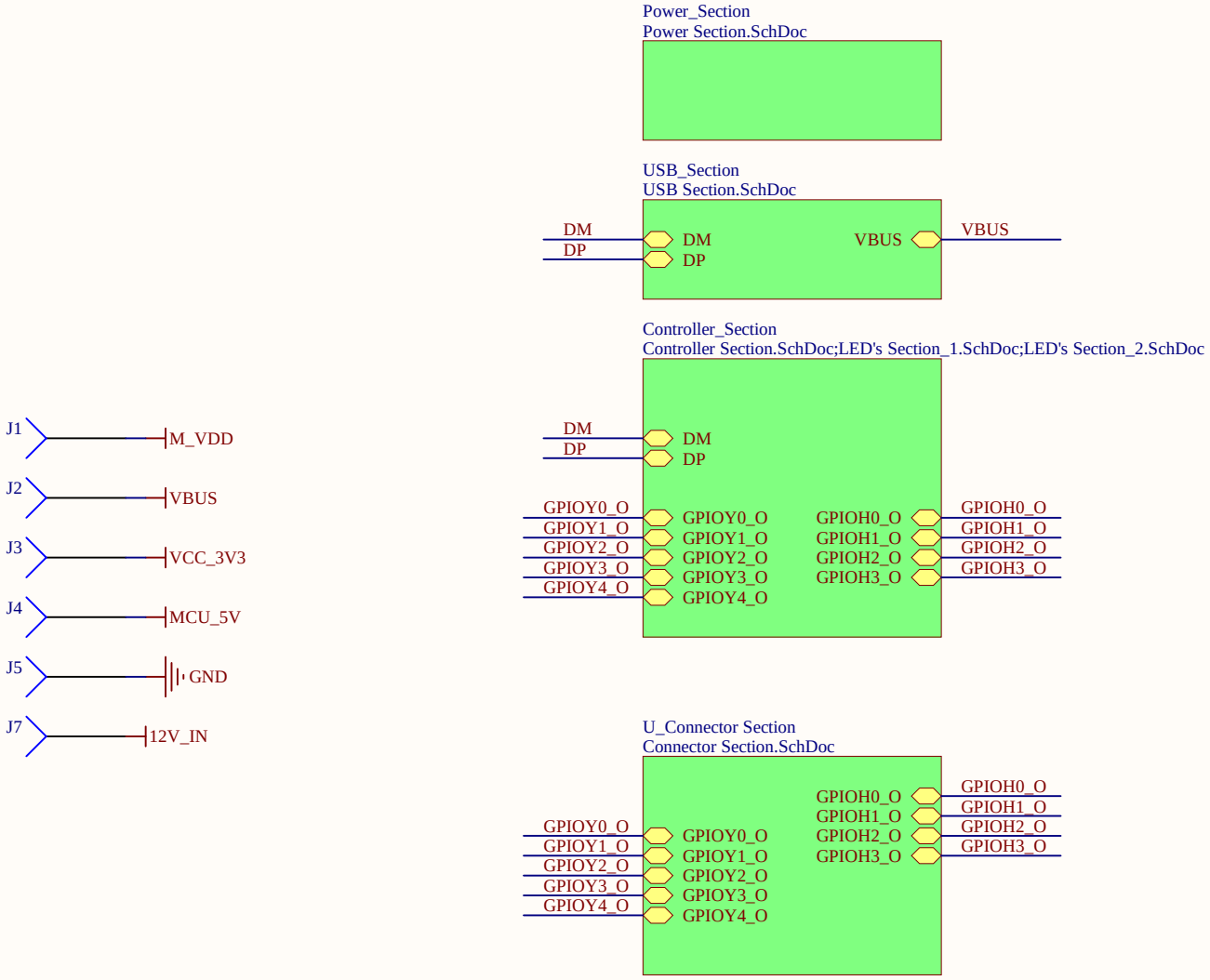
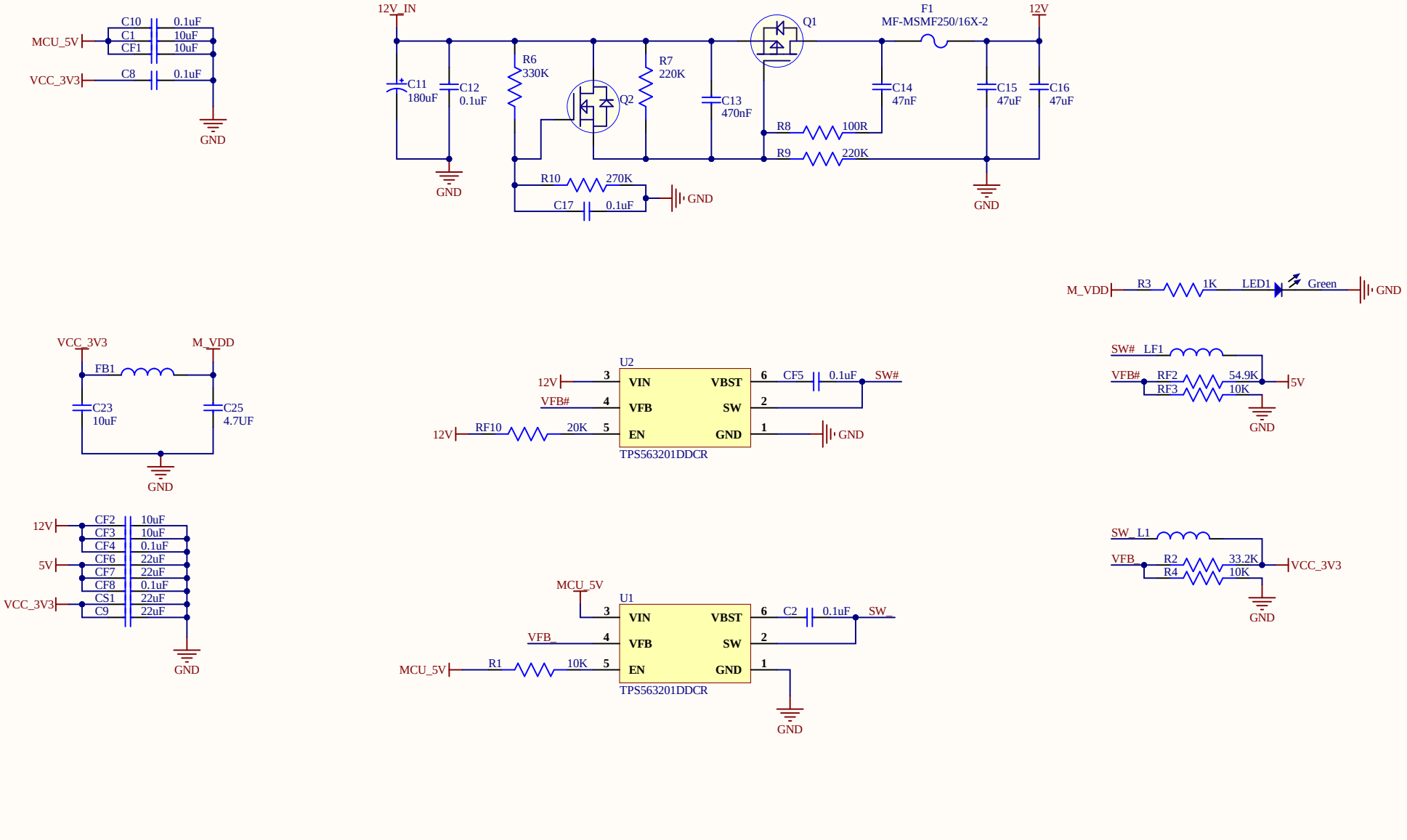
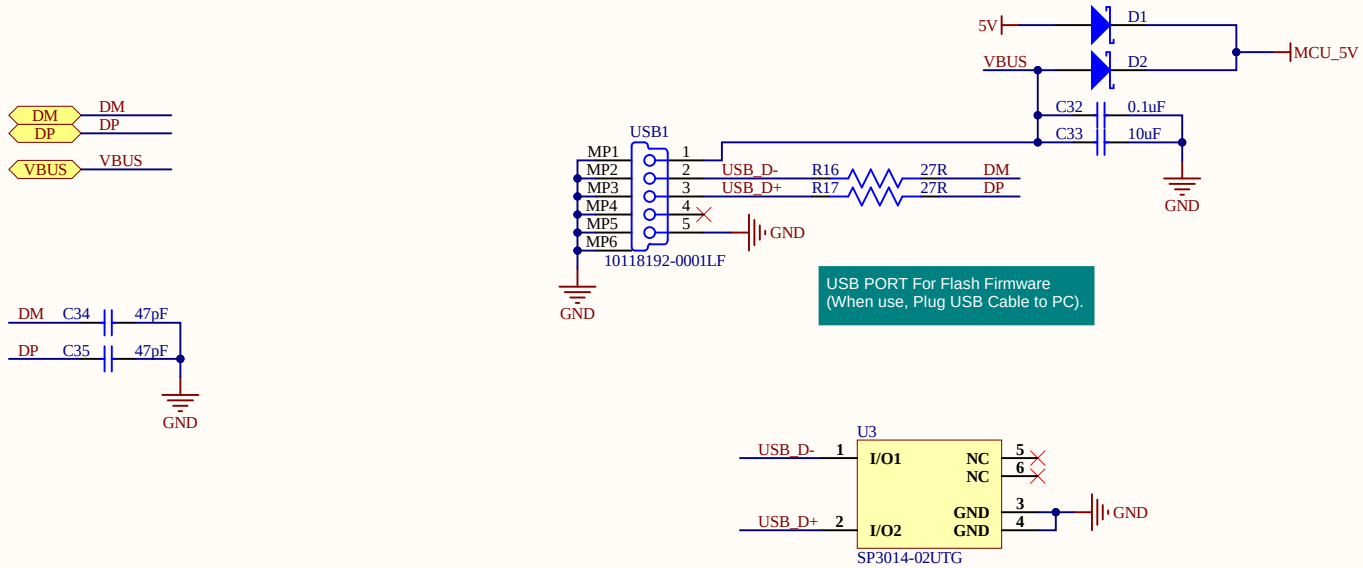
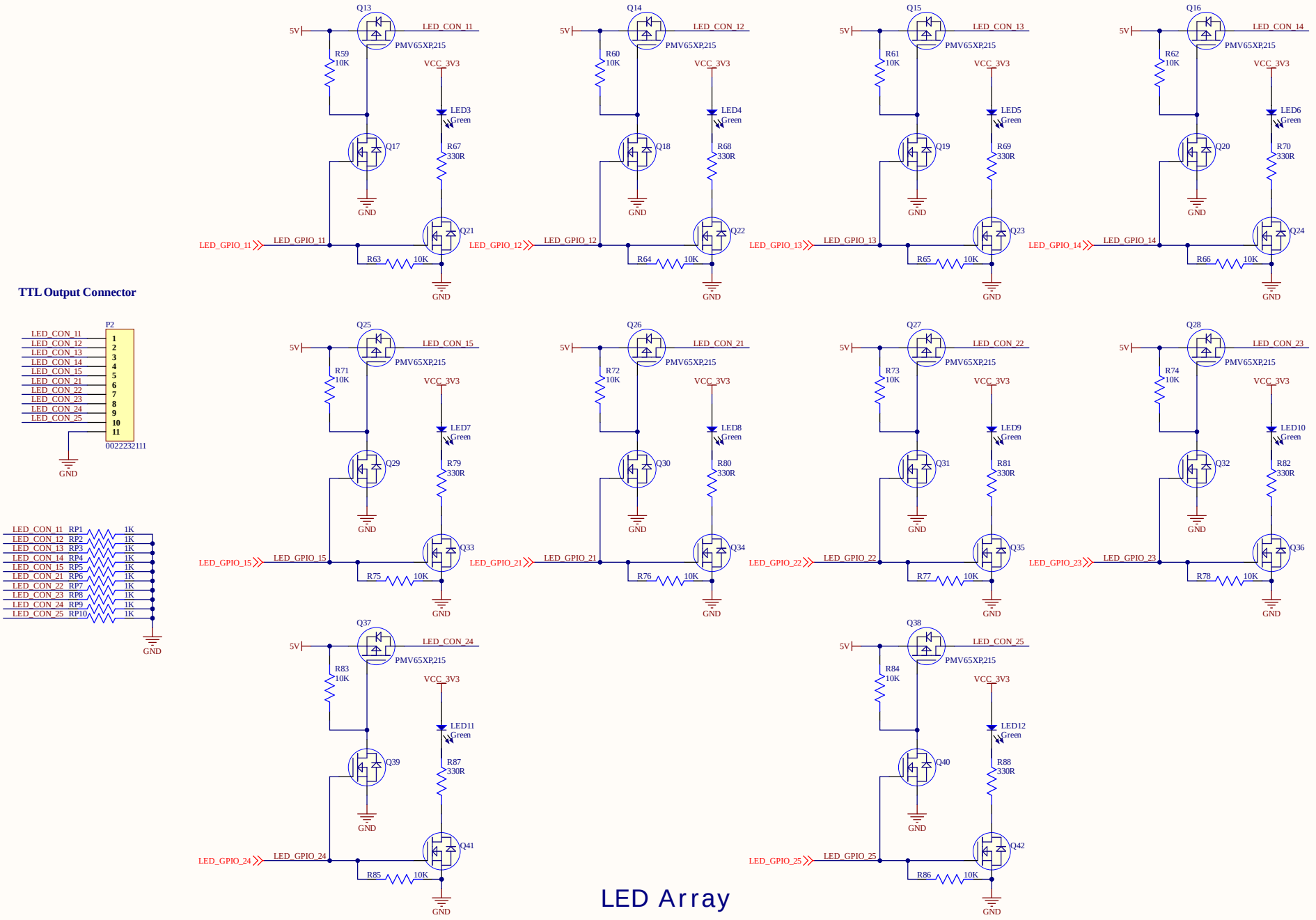

Power_Section
Power Section.SchDoc
USB_Section
USB Section.SchDoc
DM
DP
DM
DP
VBUS
VBUS
Controller_Section
Controller Section.SchDoc;LED's Section_1.SchDoc;LED's Section_2.SchDoc
DM
DP
DM
DP
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O
J1
M_VDD
J2
VBUS
J3
VCC_3V3
J4
MCU_5V
J5
GND
J7
12V_IN
U_Connector Section
Connector Section.SchDoc
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O

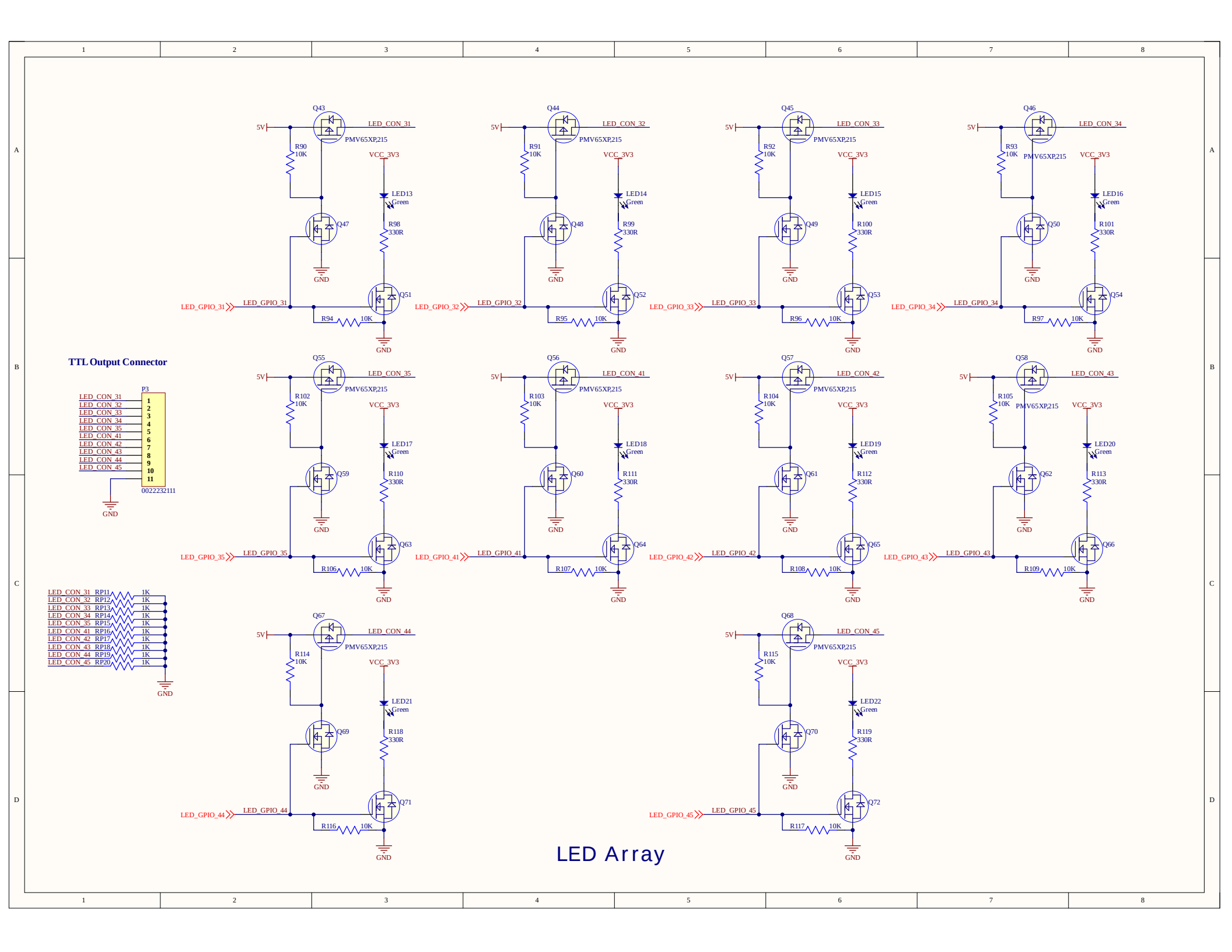


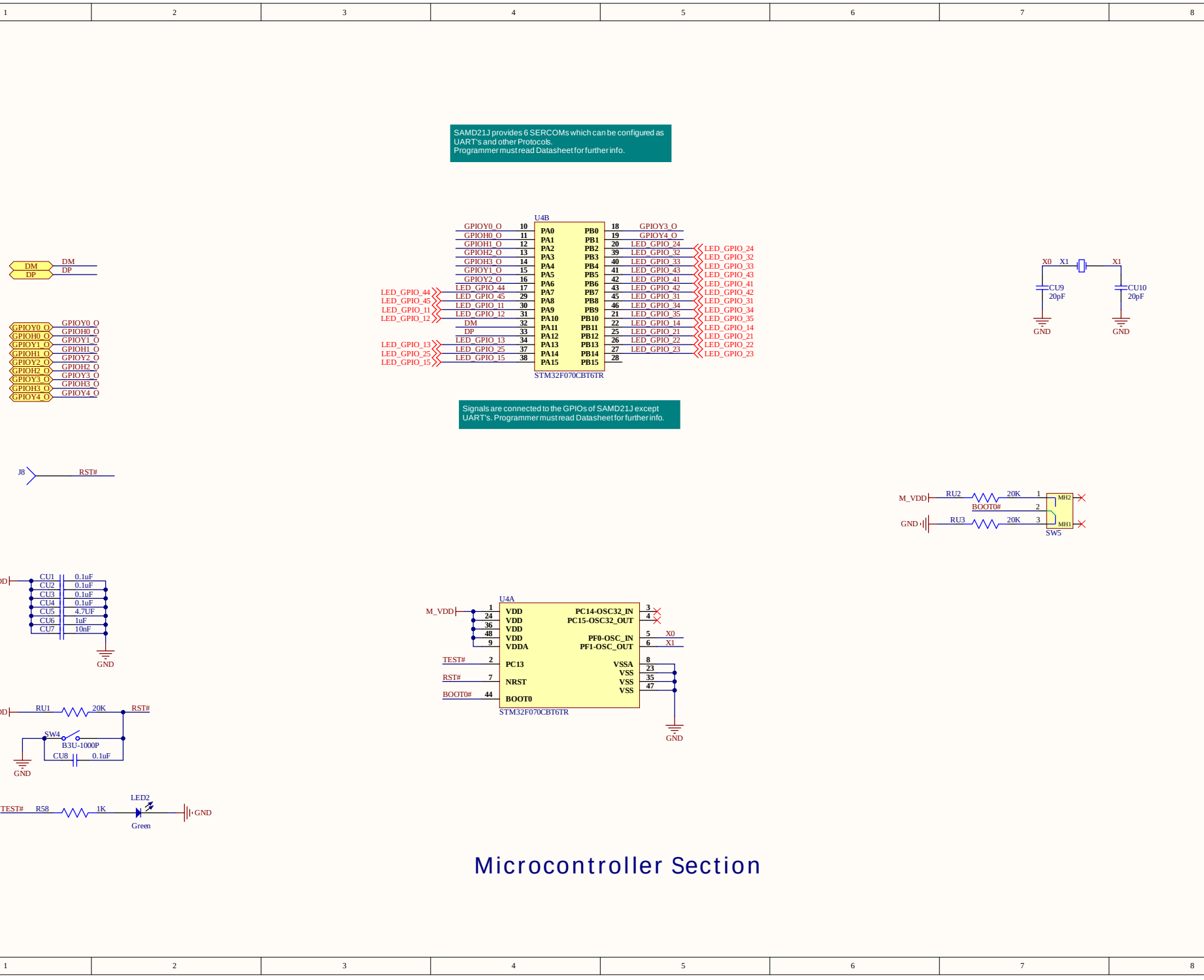
Power Section

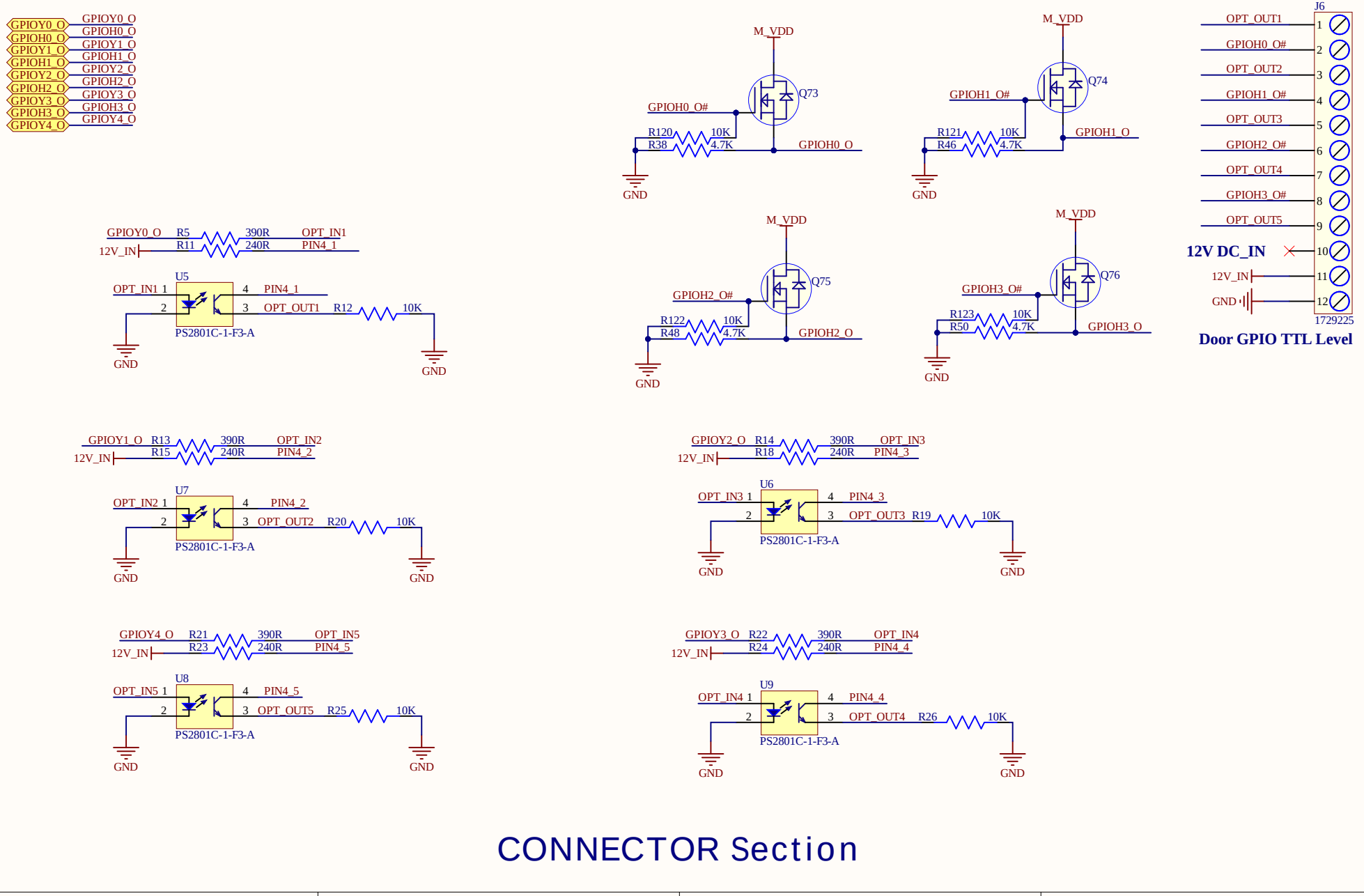

DM
DP
VBUS
DM
DP
VBUS
DM C34 47pF
DP C35 47pF
GND
5V
VBUS
D1
D2
MCU_5V
C32 0.1uF
C33 10uF
GND
USB1
MP1
MP2
MP3
MP4
MP5
MP6
1
2
3
4
5
USB_D-
USB_D+
R16 27R DM
R17 27R DP
GND
10118192-0001LF
GND
USB PORT For Flash Firmware
(When use, Plug USB Cable to PC).
U3
USB_D- 1
USB_D+ 2
I/O1
I/O2
NC 5
NC 6
GND 3
GND 4
GND
SP3014-02UTG

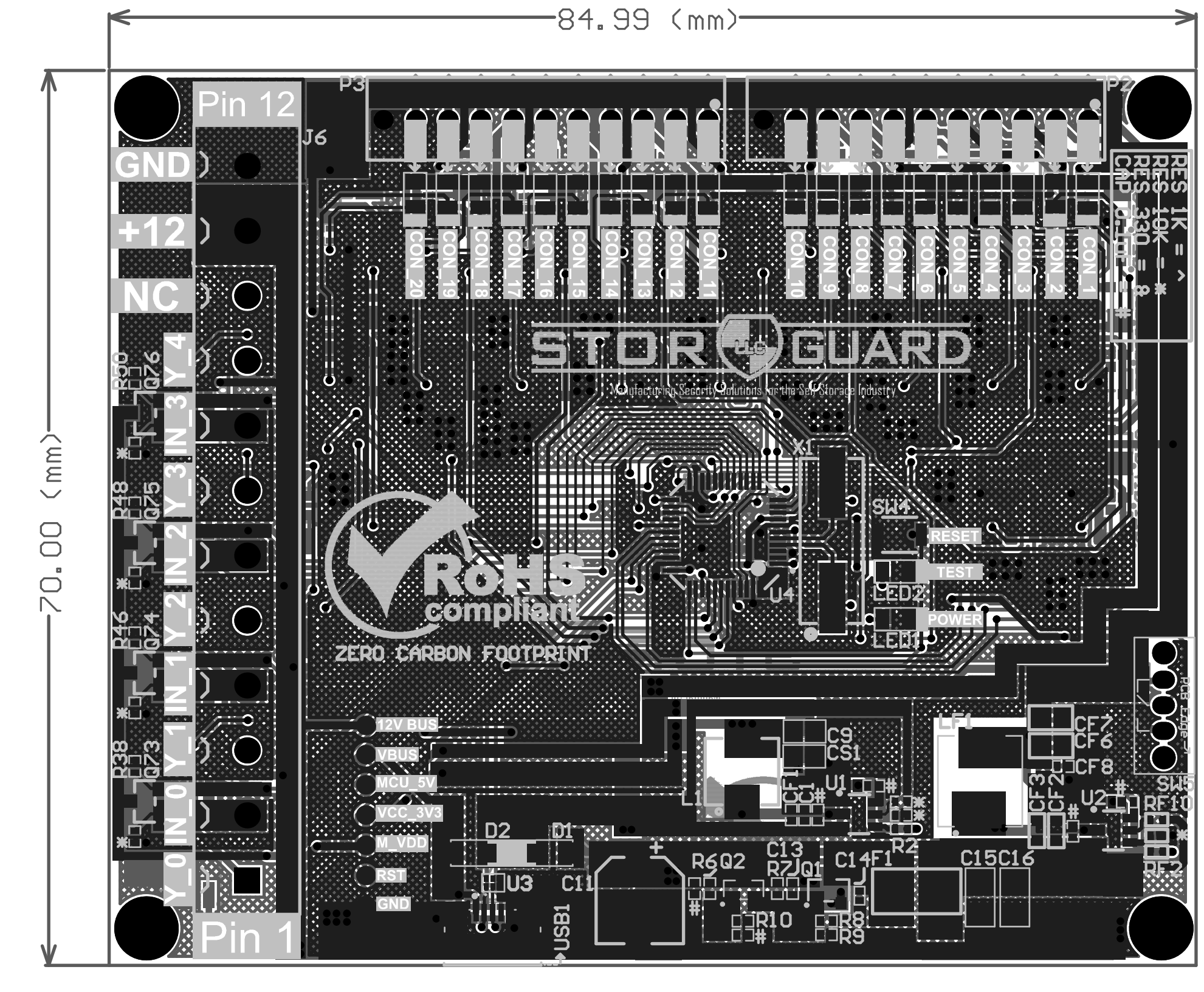
USB Section


TTL Output Connector
P2
LED_CON_11
LED_CON_12
LED_CON_13
LED_CON_14
LED_CON_15
LED_CON_21
LED_CON_22
LED_CON_23
LED_CON_24
LED_CON_25
1
2
3
4
5
6
7
8
9
10
11
0022232111
GND
LED_CON_11 RP1 1K
LED_CON_12 RP2 1K
LED_CON_13 RP3 1K
LED_CON_14 RP4 1K
LED_CON_15 RP5 1K
LED_CON_21 RP6 1K
LED_CON_22 RP7 1K
LED_CON_23 RP8 1K
LED_CON_24 RP9 1K
LED_CON_25 RP10 1K
Q13
PMV65XP,215
5V
LED_CON_11
R59
10K
VCC_3V3
LED3
Green
Q17
R67
330R
Q21
LED_GPIO_11
R63
10K
Q14
LED_CON_12
R60
LED4
Q18
R68
Q22
LED_GPIO_12
R64
Q15
LED_CON_13
R61
LED5
Q19
R69
Q23
LED_GPIO_13
R65
Q16
LED_CON_14
R62
LED6
Q20
R70
Q24
LED_GPIO_14
R66
Q25
LED_CON_15
R71
LED7
Q29
R79
Q33
LED_GPIO_15
R75
Q26
LED_CON_21
R72
LED8
Q30
R80
Q34
LED_GPIO_21
R76
Q27
LED_CON_22
R73
LED9
Q31
R81
Q35
LED_GPIO_22
R77
Q28
LED_CON_23
R74
LED10
Q32
R82
Q36
LED_GPIO_23
R78
Q37
LED_CON_24
R83
LED11
Q39
R87
Q41
LED_GPIO_24
R85
Q38
LED_CON_25
R84
LED12
Q40
R88
Q42
LED_GPIO_25
R86
LED Array


TTL Output Connector
P3
LED_CON_31
LED_CON_32
LED_CON_33
LED_CON_34
LED_CON_35
LED_CON_41
LED_CON_42
LED_CON_43
LED_CON_44
LED_CON_45
0022232111
GND
LED_CON_31 RP11 1K
LED_CON_32 RP12 1K
LED_CON_33 RP13 1K
LED_CON_34 RP14 1K
LED_CON_35 RP15 1K
LED_CON_41 RP16 1K
LED_CON_42 RP17 1K
LED_CON_43 RP18 1K
LED_CON_44 RP19 1K
LED_CON_45 RP20 1K
Q43
PMV65XP,215
5V
LED_CON_31
R90 10K
VCC_3V3
LED13 Green
Q47
R98 330R
Q51
LED_GPIO_31
LED_GPIO_31
R94 10K
Q44
LED_CON_32
R91 10K
LED14 Green
Q48
R99 330R
Q52
LED_GPIO_32
LED_GPIO_32
R95 10K
Q45
LED_CON_33
R92 10K
LED15 Green
Q49
R100 330R
Q53
LED_GPIO_33
LED_GPIO_33
R96 10K
Q46
LED_CON_34
R93 10K
LED16 Green
Q50
R101 330R
Q54
LED_GPIO_34
LED_GPIO_34
R97 10K
Q55
LED_CON_35
R102 10K
LED17 Green
Q59
R110 330R
Q63
LED_GPIO_35
LED_GPIO_35
R106 10K
Q56
LED_CON_41
R103 10K
LED18 Green
Q60
R111 330R
Q64
LED_GPIO_41
LED_GPIO_41
R107 10K
Q57
LED_CON_42
R104 10K
LED19 Green
Q61
R112 330R
Q65
LED_GPIO_42
LED_GPIO_42
R108 10K
Q58
LED_CON_43
R105 10K
LED20 Green
Q62
R113 330R
Q66
LED_GPIO_43
LED_GPIO_43
R109 10K
Q67
LED_CON_44
R114 10K
LED21 Green
Q69
R118 330R
Q71
LED_GPIO_44
LED_GPIO_44
R116 10K
Q68
LED_CON_45
R115 10K
LED22 Green
Q70
R119 330R
Q72
LED_GPIO_45
LED_GPIO_45
R117 10K
VCC
LED Array


SAMD21J provides 6 SERCOMs which can be configured as UART's and other Protocols.
Programmer must read Datasheet for further info.
U4B
PA0 PB0
PA1 PB1
PA2 PB2
PA3 PB3
PA4 PB4
PA5 PB5
PA6 PB6
PA7 PB7
PA8 PB8
PA9 PB9
PA10 PB10
PA11 PB11
PA12 PB12
PA13 PB13
PA14 PB14
PA15 PB15
STM32F070CBT6TR
Signals are connected to the GPIOs of SAMD21J except UART's. Programmer must read Datasheet for further info.
DM
DP
GPIOY0_O
GPIOH0_O
GPIOY1_O
GPIOH1_O
GPIOY2_O
GPIOH2_O
GPIOY3_O
GPIOH3_O
GPIOY4_O
J8
RST#
X0
X1
CU9 20pF
CU10 20pF
GND
M_VDD
RU2 20K
BOOT0#
RU3 20K
MH1
MH2
SW5
CU1 0.1uF
CU2 0.1uF
CU3 0.1uF
CU4 0.1uF
CU5 4.7UF
CU6 1uF
CU7 10nF
RU1 20K
SW4
B3U-1000P
CU8 0.1uF
TEST#
R58 1K
LED2
Green
U4A
VDD
VDDA
PC13
NRST
BOOT0
PC14-OSC32_IN
PC15-OSC32_OUT
PF0-OSC_IN
PF1-OSC_OUT
VSSA
VSS
Microcontroller Section


GPIOY0_O
GPIOH0_O
GPIOY1_O
GPIOH1_O
GPIOY2_O
GPIOH2_O
GPIOY3_O
GPIOH3_O
GPIOY4_O
M_VDD
Q73
GPIOH0_O#
R120
10K
R38
4.7K
GPIOH0_O
GND
M_VDD
Q74
GPIOH1_O#
R121
10K
R46
4.7K
GPIOH1_O
GND
J6
OPT_OUT1
GPIOH0_O#
OPT_OUT2
GPIOH1_O#
OPT_OUT3
GPIOH2_O#
OPT_OUT4
GPIOH3_O#
OPT_OUT5
12V DC_IN
12V_IN
GND
1729225
Door GPIO TTL Level
GPIOY0_O
R5
390R
OPT_IN1
12V_IN
R11
240R
PIN4_1
U5
OPT_IN1
PIN4_1
OPT_OUT1
R12
10K
PS2801C-1-F3-A
GND
GND
M_VDD
Q75
GPIOH2_O#
R122
10K
R48
4.7K
GPIOH2_O
GND
M_VDD
Q76
GPIOH3_O#
R123
10K
R50
4.7K
GPIOH3_O
GND
GPIOY1_O
R13
390R
OPT_IN2
12V_IN
R15
240R
PIN4_2
U7
OPT_IN2
PIN4_2
OPT_OUT2
R20
10K
PS2801C-1-F3-A
GND
GND
GPIOY2_O
R14
390R
OPT_IN3
12V_IN
R18
240R
PIN4_3
U6
OPT_IN3
PIN4_3
OPT_OUT3
R19
10K
PS2801C-1-F3-A
GND
GND
GPIOY4_O
R21
390R
OPT_IN5
12V_IN
R23
240R
PIN4_5
U8
OPT_IN5
PIN4_5
OPT_OUT5
R25
10K
PS2801C-1-F3-A
GND
GND
GPIOY3_O
R22
390R
OPT_IN4
12V_IN
R24
240R
PIN4_4
U9
OPT_IN4
PIN4_4
OPT_OUT4
R26
10K
PS2801C-1-F3-A
GND
GND
CONNECTOR Section


84.99 (mm)
70.00 (mm)
Pin 12
Pin 1
J6
GND
+12
NC
Y_0
IN_0
Y_1
IN_1
Y_2
IN_2
Y_3
IN_3
Y_4
P3
P2
CON_20
CON_19
CON_18
CON_17
CON_16
CON_15
CON_14
CON_13
CON_12
CON_11
CON_10
CON_9
CON_8
CON_7
CON_6
CON_5
CON_4
CON_3
CON_2
CON_1
RES 1K = >
RES 10K = *
RES 330 = #
CAP 0.1U = #
STOR GUARD
Manufacturing Security Solutions for the Self Storage Industry
RoHS compliant
ZERO CARBON FOOTPRINT
X1
SW4
RESET
TEST
POWER
LED2
LED1
U4
12V BUS
VBUS
MCU_5V
VCC_3V3
M_VDD
RST
GND
D2
D1
U3
C11
USB1
R6
Q2
C13
R7
Q1
C14
F1
R2
C15
C16
R10
R8
R9
C9
CS1
CF1
C1
U1
L1
LF1
CF7
CF6
CF8
CF3
CF2
U2
SW5
RF1
RF2
R50
Q76
R48
Q75
R46
Q74
R38
Q73